SHERYIANS AI SCHOOL — ADVANCED STATISTICS NOTES

Focused Study Sheet: ANOVA, Covariance, and Correlation

1. Covariance

Covariance evaluates how two quantitative variables change together. If both variables tend to increase or decrease simultaneously, the covariance is positive; if one increases while the other decreases, it is negative.

$$\text{Cov}(X, Y) = \sum (X_i - \bar{X})(Y_i - \bar{Y}) / (n - 1)$$

- **Direction Only:** Indicates whether variables move in the same or opposite directions.
- **Scale Dependency:** Range is unbounded ($-\infty$ to $+\infty$). Because the metric depends on variable measurement units, the magnitude cannot be used to gauge the strength of the relationship.

2. Correlation (Pearson's r)

Correlation standardizes covariance to quantify both the direction and the linear strength of the association between two continuous variables.

$$r = \text{Cov}(X, Y) / (\sigma_X \times \sigma_Y)$$

- **Bounded Scale:** Always falls between −1.0 and +1.0 regardless of units.
- **r = +1.0:** Perfect positive linear relationship.
- **r = 0.0:** No linear relationship between variables.
- **r = −1.0:** Perfect negative linear relationship.

3. ANOVA (Analysis of Variance)

ANOVA tests for statistical differences among the means of **three or more independent groups** simultaneously.

- **Purpose:** Prevents Type I Error (false positive) inflation that occurs when conducting multiple pairwise t-tests (which compounds $\alpha = 0.05$ across comparisons).
- **Null Hypothesis (H_0):** $\mu_1 = \mu_2 = \mu_3 = ... = \mu_k$ (All group population means are equal).
- **Alternative Hypothesis (H_a):** At least one group mean is significantly different from the others.
- **Test Statistic (F-Ratio):**

  $$F = \text{Variance Between Groups} / \text{Variance Within Groups}$$

- **Decision Rule:** If F is large and the p-value $\leq \alpha$ (0.05), reject H_0 and conclude that at least one group mean differs.

Key takeaway for GitHub log: Covariance provides movement direction; Correlation standardizes this into strength from −1 to +1; ANOVA expands hypothesis testing to evaluate differences across 3+ group means without compounding error rates.